

A Minimal Copper-Plate Storage-Scheduling Benchmark for Differential Dynamic Programming

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I. COPPER-PLATE STORAGE-SCHEDULING MODEL

A. Optimization Problem

A single storage device and a single grid connection serve a known demand p_L^t over T periods of length Δt , all quantities in per unit. Following the sign convention of [1], $P_B^t > 0$ discharges and $P_B^t < 0$ charges; B^t is the energy at the *start* of period t , so B^1 is given and B^{T+1} is terminal. With $t \in \mathcal{T} = \{1, \dots, T\}$:

$$\min_{P_{\text{Subs}}, P_B, B} \sum_{t=1}^T \left[c^t P_{\text{Subs}}^t + C_B (P_B^t)^2 \right] \Delta t + w (B^{T+1} - B^1)^2 \quad (1a)$$

$$\text{s.t. } P_{\text{Subs}}^t + P_B^t - p_L^t = 0, \quad t \in \mathcal{T} \quad (1b)$$

$$B^1 = B_0 \quad (1c)$$

$$B^{t+1} = B^t - P_B^t \Delta t, \quad t \in \mathcal{T} \quad (1d)$$

$$P_{\text{Subs}}^t \geq 0, \quad t \in \mathcal{T} \quad (1e)$$

$$P_B^{\min} \leq P_B^t \leq P_B^{\max}, \quad t \in \mathcal{T} \quad (1f)$$

$$B^{\min} \leq B^{t+1} \leq B^{\max}, \quad t \in \mathcal{T} \quad (1g)$$

The *base instance* is (1a)–(1d), with all three bounds dropped.

B. Optimal-Control Form

Restated over $N = T$ stages in the form required by DDP:

$$\begin{aligned} \min_{x, u} \quad & \sum_{t=1}^{N-1} \ell(x^t, u^t) + \ell^F(x^N, u^N) \\ \text{s.t. } \quad & x^1 = \hat{x}^1, \\ & x^{t+1} = f(x^t, u^t), \quad t \in \{1, \dots, N-1\}, \\ & c(x^t, u^t) = 0, \quad t \in \{1, \dots, N\}. \end{aligned} \quad (2)$$

with state, control and initial condition

$$x^t = (B^t, \tau^t)^\top \in \mathbb{R}^2, \quad u^t = (P_{\text{Subs}}^t, P_B^t)^\top \in \mathbb{R}^2, \quad (3)$$

$\hat{x}^1 = (B_0, 1)^\top$. Period-varying data is carried by the time index τ through the Lagrange interpolants on the nodes $\tau = 1, \dots, T$,

$$p_L(\tau) = \sum_{k=1}^T p_L^k \prod_{\substack{m=1 \\ m \neq k}}^T \frac{\tau - m}{k - m}, \quad (4)$$

and likewise for the price $c(\tau)$. The stage data is then

$$f(x, u) = (x_1 - u_2 \Delta t, x_2 + 1)^\top, \quad (5)$$

$$c(x, u) = u_1 + u_2 - p_L(x_2), \quad (6)$$

$$\ell(x, u) = [c(x_2) u_1 + C_B u_2^2] \Delta t, \quad (7)$$

$$\begin{aligned} \ell^F(x, u) = & [c^T u_1 + C_B u_2^2] \Delta t \\ & + w (x_1 - u_2 \Delta t - B_0)^2. \end{aligned} \quad (8)$$

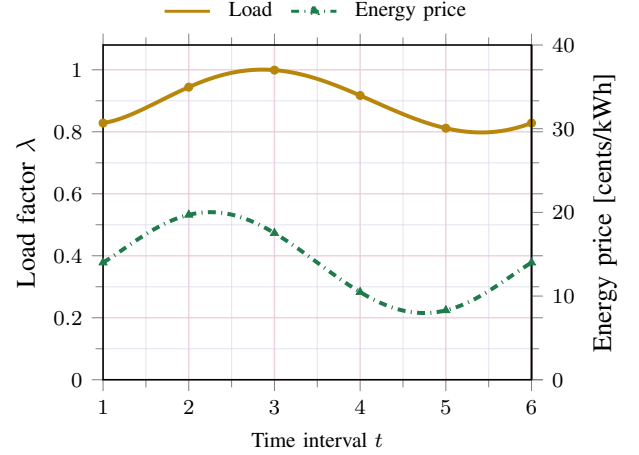


Fig. 1. Input profiles at $T = 6$: load factor (left axis), energy price (right). Markers are the data; the curves are the Lagrange interpolants, which is what the solver actually evaluates. Overshoot between nodes is slight (1.0005 against a node maximum of 0.999), but the interpolant degree is $T - 1$.

Bounds (1e)–(1f) map onto control limits $b^L \leq u^t \leq b^U$. The state bound (1g) is appended as a second row of c via a slack s^t adjoined to the control, giving $n_c = 2$, $n_u = 3$:

$$(x_1 - u_2 \Delta t) - B^{\min} - s^t = 0, \quad 0 \leq s^t \leq B^{\max} - B^{\min}. \quad (9)$$

C. Closed-Form Reference

The base instance is a strictly convex equality-constrained QP. Eliminating $P_{\text{Subs}}^t = p_L^t - P_B^t$ and $B^{T+1} - B^1 = -\Delta t \sum_t P_B^t$ reduces it to

$$\begin{aligned} J(P_B) = & \text{const} - \Delta t \sum_t c^t P_B^t + C_B \Delta t \sum_t (P_B^t)^2 \\ & + w (\Delta t)^2 s^2, \end{aligned} \quad (10)$$

with $s = \sum_t P_B^t$. Stationarity divided through by Δt is $-c^k + 2C_B P_B^k + 2w \Delta t s = 0$; summing over k collapses it to

$$s = \frac{\sum_t c^t}{2C_B + 2Tw \Delta t}, \quad P_B^k = \frac{c^k - 2w \Delta t s}{2C_B}. \quad (11)$$

REFERENCES

- [1] A. R. Jha, S. Paul, and A. Dubey, "Analyzing the performance of linear and nonlinear multi-period optimal power flow models for active distribution networks," in *2025 IEEE North-East India International Energy Conversion Conference and Exhibition (NE-IECCCE)*. IEEE, 2025, pp. 04–06.

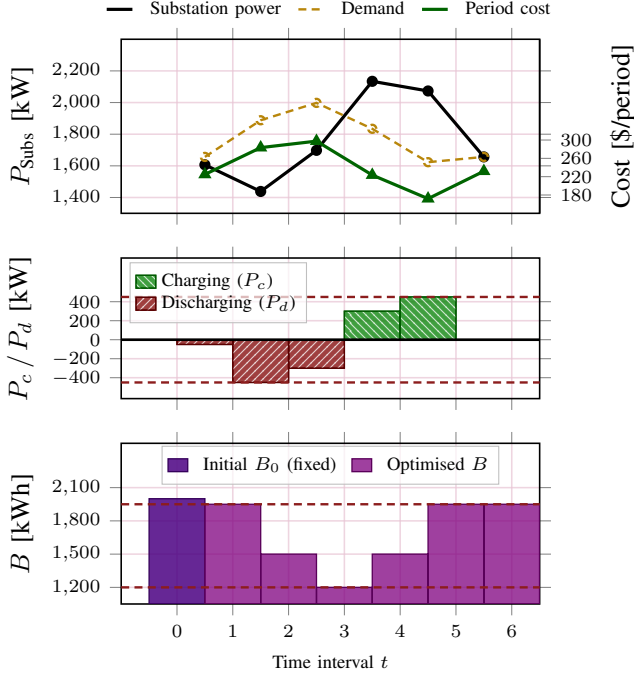


Fig. 2. The benchmark at $T = 6$ with all bounds imposed. Charging is drawn upward, discharging downward, so a bar's direction is the device's action rather than the sign of P_B^t . The battery discharges through $t = 1$ –3 and recharges at $t = 4, 5$, riding its ± 450 kW rails and both energy limits (dashed). Those five bounds are the active set the reference solver reports.

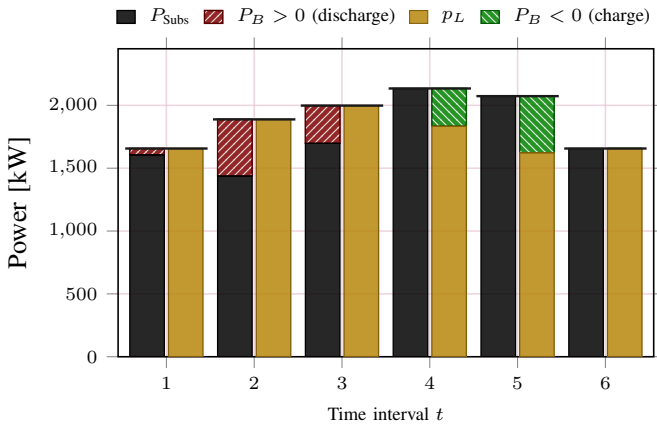


Fig. 3. Power-balance composition at $T = 6$. Per interval, supply (P_{Subs}^t plus discharge) on the left against demand (p_L^t plus charge) on the right; the equal heights are the power balance, marked by the rule. At $t = 2$ the battery supplies 23.8% of the node's 1888 kW; at $t = 5$ its charging instead raises import to 2073 kW to serve 1623 kW of load.

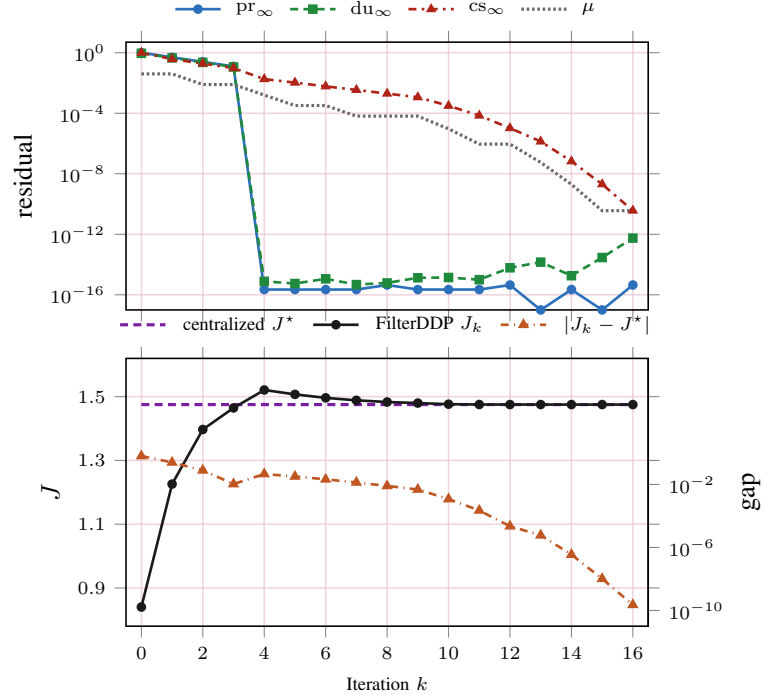


Fig. 4. FilterDDP convergence with all bounds imposed, 17 iterations. Top: the KKT residuals and the barrier parameter. Feasibility is reached almost immediately — pr_∞ and du_∞ hit machine precision by $k = 4$ — after which progress is entirely μ driving cs_∞ down alongside it, so the iteration count is set by the barrier schedule rather than by the optimisation. Bottom: the objective against the centralised JuMP/Ipopt baseline J^* , with the gap on the right axis. J_k is not monotone — it overshoots J^* near $k = 5$ and returns from above, because the line search decreases the barrier objective rather than J — and the gap closes to 2.3×10^{-10} .

TABLE I
FILTERDDP ITERATIONS, ALL BOUNDS

k	objective	backward pass				forward pass	
		pr_∞	du_∞	cs_∞	$\log_{10} \mu$	α	bk
0	0.840000	$1.0e+00$	$9.2e-01$	$1.0e+00$	-1.40	0.50	1
1	1.225783	$5.0e-01$	$4.6e-01$	$3.8e-01$	-1.40	0.50	1
2	1.397002	$2.5e-01$	$2.3e-01$	$1.9e-01$	-2.10	0.50	1
3	1.464625	$1.2e-01$	$1.1e-01$	$9.6e-02$	-2.10	1.00	0
4	1.520935	$2.2e-16$	$7.8e-16$	$1.8e-02$	-2.80	0.50	1
5	1.507179	$2.2e-16$	$5.6e-16$	$1.1e-02$	-3.49	0.50	1
6	1.496237	$2.2e-16$	$1.1e-15$	$6.0e-03$	-3.49	0.50	1
7	1.488496	$2.2e-16$	$4.7e-16$	$3.5e-03$	-4.19	0.50	1
8	1.483187	$4.4e-16$	$6.1e-16$	$2.0e-03$	-4.19	0.50	1
9	1.479927	$2.2e-16$	$1.3e-15$	$1.1e-03$	-4.19	1.00	0
10	1.476349	$2.2e-16$	$1.4e-15$	$3.1e-04$	-5.03	1.00	0
11	1.475395	$2.2e-16$	$1.0e-15$	$7.0e-05$	-6.04	1.00	0
12	1.475195	$4.4e-16$	$6.1e-15$	$1.0e-05$	-6.04	1.00	0
13	1.475179	$0.0e+00$	$1.4e-14$	$1.4e-06$	-7.25	1.00	0
14	1.475173	$2.2e-16$	$1.8e-15$	$6.7e-08$	-8.70	1.00	0
15	1.475173	$0.0e+00$	$2.9e-14$	$2.0e-09$	-10.44	1.00	0
16	1.475173	$4.4e-16$	$5.6e-13$	$3.7e-11$	-10.44	—	—

The backward pass evaluates the residuals at the current iterate; the forward pass is the filter line search that follows it, with accepted step α and $\text{bk} = \log_2(1/\alpha)$ backtracks. FilterDDP prints these two halves one iteration out of step, because `iteration_status` is called before `forward_pass!`; the rows here are realigned so each holds one backward pass and the forward pass it actually fed. The last row has no forward pass — the solver converged instead of stepping. Regularisation never fired. Every backtrack in the table was a rollout rejection rather than a filter or sufficient-decrease rejection, and the solver's own `ls` counter records none of them.

TABLE II
INSTANCE DATA

Interval t	1	2	3	4	5	6
Demand p_L^t	1.657	1.888	1.998	1.834	1.623	1.657
Energy price c^t	0.140	0.197	0.175	0.105	0.083	0.140
Period length Δt	1	Initial energy B_0		2.0		
Throughput C_B	0.05	Terminal weight w		5.0		
<i>Bounds, constrained variants only</i>						
$P_B^{\min}, P_B^{\max}$				−0.45, 0.45		
$B^{\min}, B^{\max}$				1.20, 1.95		

All quantities are in per unit on the tADMM base, 1 p.u. = 1000 kW and 1 p.u.h = 1000 kWh; with $\Delta t = 1$, power and energy are numerically interchangeable, and the bounds are ± 450 kW and 1200–1950 kWh.

Figs. 2 and 3 report battery quantities in kW and kWh rather than per unit. The series are the tADMM load and price profiles, $p_L^t = 2 \lambda^t$ with $\lambda^t = 0.8 + 0.1(\sin(\theta^t - 0.8) + 1)$ and $c^t = 0.08 + 0.06(\sin \theta^t + 1)$, sampled at $\theta^t \in \text{range}(0, 2\pi, T)$. These *resample* with T , so a shorter horizon is not a prefix of this row; both endpoints of $[0, 2\pi]$ are included, which is why $c^1 = c^T$. The -0.8 phase offset on the load is what keeps the two series from being collinear; their Pearson correlation is 0.644. Bounds apply only to the constrained variants.